

Fast Location Survey of DGNSS Reference Station to Support UAV Navigation

Dong-Kyeong Lee¹, Minchan Kim¹, Kiwan Kim¹ and Jiyun Lee^{1*}

¹ Department of Aerospace Engineering, Korea Advanced Institute of Science and Technology,
Daejeon, 305-701, Korea (jiyunlee@kaist.ac.kr) * Corresponding author

Abstract: Mobile Differential Global Navigation Satellite Systems (Mobile DGNSS) formed with multiple UAVs can support missions on a battlefield by providing navigation with high accuracy and integrity. For time efficiency and maneuverability of the system, the reference stations that are deployed and sited on the battlefield should be able to determine their precise position rapidly before generating the differential corrections. This paper presents a methodology for fast location surveying of the reference stations, which improves the GNSS-based position estimates of reference stations using precise measurements of relative distances between the stations. Simulation results showed that the proposed method enables users to achieve the desired accuracy in less time than the simple method which averages stand-alone GNSS solutions only. A theoretical model of DGNSS position uncertainty induced by inaccurate surveyed locations of the reference stations was also defined in this paper. The model was validated by comparing it to the results from experiments conducted with a DGNSS test-bed equipped with three GNSS receivers and a pseudo-user.

Keywords: Mobile DGNSS, Surveyed Location, Precise Relative Distance.

1. INTRODUCTION

One of the most essential technologies required for the operation of unmanned aerial vehicles (UAVs) is the reliable and accurate computation of navigation information. As most UAVs obtain their navigation information from onboard GNSS sensors, it is imperative to research a way to improve GNSS position accuracy. High accuracy can be obtained through the development of DGNSS technology, which uses differential correction provided by a reference station sited at a fixed location.

Fig. 1 illustrates the concept of mobile DGNSS whose reference stations can move, site at an arbitrary location within service area, and provide differential corrections to GNSS users who carry out missions on a battlefield. The mobile DGNSS is composed of three or more UAVs equipped with GNSS receivers and a RF data broadcast transmitter responsible for sending differential corrections to the users. Using the GNSS measurements obtained from the reference receiver, the mobile DGNSS monitors any system failures which may pose integrity risks to the users, and it also generates accurate and reliable correction. However, DGNSS correction uncertainties induced by inaccurate surveyed locations of the reference stations degrade the position solutions calculated by the DGNSS user. Therefore, when mobile DGNSS operates in unpredictable environments such as battlefields, it is essential to determine accurate surveyed positions of reference stations in a short period of time to ensure satisfactory performance.

This paper develops a methodology for fast location surveying of the reference stations. The core of this methodology lies in the correction of stand-alone GNSS-based positions of reference stations using data quality assessment and accurate measurements of

relative distances between the stations. The quality of data can be determined from factors such as dilution of precision (DOP) and carrier-to-noise density (C/N₀). The high accuracy of the relative distances can be attained by using precise carrier phase measurements obtained from GNSS receivers. Even if this method requires some computation load, it is effective in minimizing the payload of the aircrafts, since sensors other than the GNSS receivers, such as radar and lidar, are not required onboard to obtain precise relative distances.

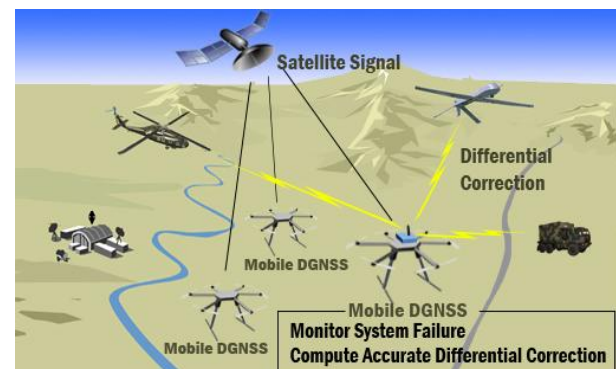


Fig. 1 Mobile DGNSS Concept

This paper analyzes the rate of surveyed position convergence to the true position. In addition, a theoretical model of DGNSS position errors induced by inaccurate locations of reference stations is defined. The proposed model was validated by experiments with a mobile DGNSS test-bed equipped with three GNSS receivers and a pseudo-user. The data obtained from the test-bed was also used to test the applicability of fast location survey algorithm proposed in this paper. The validity of the algorithm itself was confirmed using a simulation.

2. SURVEYED LOCATION ERROR MODEL

DGNSS generates accurate differential corrections using the GNSS data set obtained from multiple reference receivers whose location are known. To compute the differential correction (ρ_{corr}), firstly a correction ($\rho_{c,m}^n$) for each Receiver m and Satellite n pair is calculated as shown Eq. (1).

$$\rho_{c,m}^n = \rho_m^n - R_m^n + \tau_m^n \quad (1)$$

Where τ_m^n is the clock correction, ρ_m^n is the pseudorange measurement, and R_m^n is the range from the reference antenna to the satellite. The calculations are based on the known locations of the reference antennas and the broadcast ephemeris. Secondly, the receiver clock biases are adjusted as outlined in Eq. (2).

$$\rho_{ca,m}^n = \rho_{c,m}^n - \frac{1}{N} \sum_{j \in S_c} \rho_{c,m}^j \quad (2)$$

In this equation, S_c is the set of N satellites observed at each epoch. For the next step, the averaged corrections for each satellite are computed as follows in Eq. (3).

$$\rho_{corr}^n = \frac{1}{M} \sum_{i \in S_n} \rho_{ca,i}^n \quad (3)$$

Here, S_n is the set of M reference receivers. The ultimately computed ρ_{corr} is broadcast to users within 10 to 200 km vicinity of the reference station.

The surveyed location error (δX_e) of the reference station due to the insufficient number of location samples, incurs the pseudorange error (δR_e) in the range term between the reference antenna and the satellite. In this case, differential correction (ρ_{corr_error}) is derived as outlined in Eqs. (4) ~ (5).

$$\rho_{c,m}^n = \rho_m^n - (R_m^n + \delta R_{e,m}^n) + \tau_m^n \quad (4)$$

$$\rho_{corr_error}^n = \frac{1}{M} \sum_{i \in S_n} \rho_{ca,i}^n - \frac{1}{M} \sum_{i \in S_n} \delta R_{e,i}^n \quad (5)$$

If the DGNSS user applies this erroneous correction, the differentially corrected pseudorange measurement of the user is represented as in Eq. (6).

$$\begin{aligned} \rho_{corrected}^n &= \rho_u^n - \rho_{corr_error}^n \\ &= R^n + c(\delta t_u) + \frac{1}{M} \sum_{i \in S_n} \delta R_{e,i}^n \end{aligned} \quad (6)$$

Where c is the speed of light in vacuum state and δt_u is the receiver clock bias of the user. The relationship between the estimated position of the user ($\hat{X}_u$) and the initial estimation (X_0) can be written as in Eq. (7).

$$\hat{X}_u = X_0 + \delta X_u \quad (7)$$

If we calculate δX_u using the corrupted pseudorange measurements from Eq. (6), the errors induced by the inaccurate locations of the reference stations are derived as in Eq. (8).

$$\begin{aligned} \delta X_u &= G^{-1} \delta \rho_{corrected} = G^{-1} \left[\delta \rho_{true} + \frac{1}{M} \sum_{i \in S_n} \delta R_{e,i} \right] \\ &= \delta X_{true} + \frac{1}{M} \sum_{i \in S_n} \delta X_{e,i} \end{aligned} \quad (8)$$

In this equation, δX_{true} is the deviation of the initial estimation position from the true position. As shown in Eq. (8), the DGNSS surveyed location error model depends on the number of receivers and the sum of errors from inaccurate locations.

3. METHOD

3.1 Selective positioning using data quality

The algorithm takes GNSS position measurements as inputs to produce more accurate location measurements. Due to various error factors, the quality of GNSS position measurements may be significantly degraded. Therefore, in order to alleviate the influence of the outlier data on the surveyed location determination, a threshold is established. The threshold is an arbitrary value selected from a numeric scale which represents the quality of data. The quality of data can be determined from parameters such as DOP or S/N0 of each GNSS measurement. When the data quality of a measurement fails to meet the threshold, the current quality is compared to that of the previous measurement. If the earlier measurement has better quality, the newly acquired GNSS position measurement is discarded and replaced with the previous one.

3.2 Accurate relative distance estimation

The next part of the fast location surveying method proposed in this paper requires highly accurate relative distances between the reference stations. These distances can be obtained by using carrier phase measurements from GNSS receivers. If the reference stations of mobile DGNSS are within a few tens of kilometers proximity, centimeter-level relative positioning can be achieved by employing the carrier phase double difference observables ($\nabla \Delta \phi^{ij}$).

$$\lambda \cdot \phi^i = R^i + c \cdot (\delta t_u - \delta t^i) - I^i + T^i + \lambda \cdot N^i + \varepsilon_\phi^i \quad (9)$$

$$\lambda \cdot \nabla \Delta \phi^{ij} = \nabla \Delta R^{ij} - \nabla \Delta I^{ij} + \nabla \Delta T^{ij} + \lambda \cdot \nabla \Delta N^{ij} + \varepsilon_{\nabla \Delta \phi}^{ij} \quad (10)$$

Where λ is the wavelength of the carrier. The carrier phase measurement of satellite i is composed of geometric range (R^i), integer ambiguity (N^i), and errors which include: satellite clock bias (δt^i), receiver clock bias (δt_u), ionospheric and tropospheric delay (I^i, T^i), and other errors (ε_ϕ^i) as shown in Eq. (9). In order to eliminate the aforementioned errors, firstly the carrier phase measurements between two stations for a given satellite are subtracted, which is known as single differencing. Secondly double differences are calculated as in Eq. (10) by differencing the single differences

between two given satellites [1]. Eq. (10) describes the process of cancelling out the satellite and receiver clock biases. Moreover, if the reference stations are not too far away from one another, double differences of ionospheric and tropospheric delays can be considered as zero. As a result, double difference observables can be represented as in Eq. (11).

$$\lambda \nabla \Delta \phi^{ij} = \nabla \Delta R^{ij} + \lambda \nabla \Delta N^{ij} + \varepsilon_{\nabla \Delta \phi}^{ij} \quad (11)$$

In order to estimate the integer ambiguities and relative positions, additional observables are needed. Therefore, the geometry-free observable (Z_{GF}) is included.

$$Z_{GF} = (\phi_{L1} - \phi_{L2}) - \left(\frac{PR_{L1}}{\lambda_{L1}} + \frac{PR_{L2}}{\lambda_{L2}} \right) \left(\frac{\lambda_{L2} - \lambda_{L1}}{\lambda_{L1} + \lambda_{L2}} \right) \quad (12)$$

The geometry-free observable is calculated with L1/L2 code measurement (PR) and carrier phase measurement (ϕ) following Eq. (12). The double differences of the geometry-free observables are also computed in the same way mentioned above. If the double differences of L1 carrier, L2 carrier, and geometry-free observables are given, it is possible to estimate the relative distances ($\mathbf{x}_{Rel}$) and integer ambiguity differences ($\nabla \Delta N$) by applying the least-square method [2].

$$\begin{bmatrix} \nabla \Delta Z_{GF} \\ \nabla \Delta \phi_{L1} \\ \nabla \Delta \phi_{L2} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ -\Delta \mathbf{e}^T \\ -\Delta \mathbf{e}^T \end{bmatrix} \mathbf{x}_{Rel} + \begin{bmatrix} \mathbf{I} & -\mathbf{I} \\ \lambda_{L1} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \lambda_{L2} \mathbf{I} \end{bmatrix} \begin{bmatrix} \nabla \Delta N_{L1} \\ \nabla \Delta N_{L2} \end{bmatrix} + \varepsilon \quad (13)$$

In Eq. (13), geometry-difference ($\Delta \mathbf{e}^T$) represents the difference between the line-of-sight vectors for the master satellite and the slave satellite [2].

As the least-square method can only provide rational integer ambiguities, an additional process is required to constrain the ambiguities as integers. To deal with this problem, the well-known Least-squares Ambiguity Decorrelation Adjustment (LAMBDA) method is applied [3, 4].

3.3 Repositioning with relative distances

This part of the fast location surveying algorithm relies on two conditions. Firstly, there are at least three GNSS receivers available. Secondly, the accuracy of the relative distances between receivers is high enough, so that their errors are negligible compared to the GNSS position errors of the receivers.

The role of this part of the algorithm is to distribute the initial GNSS errors while satisfying the inter-distance measurements between the receivers, which will enable the user to obtain a new set of GNSS positions that is closer to the true position. The fact that the estimated positions are closer to the true position leads to the advantage of determining an accurate surveyed position in a shorter period of time than currently achievable.

Let us define high error correlation as the case when the position errors for each receiver are almost the same, while low error correlation is when the magnitude and

the direction of the errors for each receiver are different. Due to the nature of the algorithm, the lower the error correlation, the zero mean sum of the errors will be closer to zero, leading to more accurate estimates. Conversely, if the correlation is high, the accuracy of the estimates are questionable.

Overall, this part of the method consists of two sections. The first section is responsible for finding a new set of positions that satisfy the inter-distance measurements between the receivers. The second section is responsible for finding the set of positions for which the error of the new estimates from the GNSS positions is minimal while the inter-distance measurements are satisfied. The method is illustrated in Fig. 2.

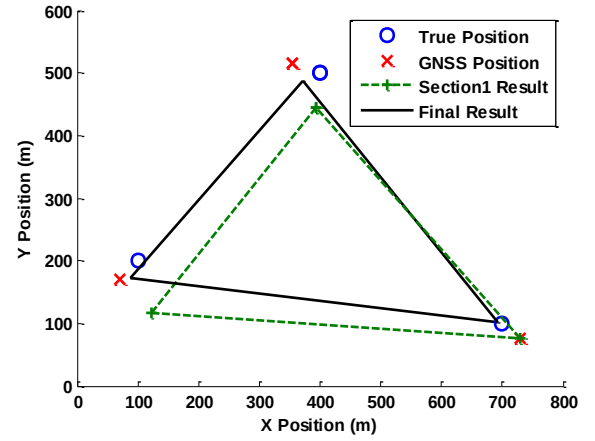


Fig. 2 Overview of Fast Location Surveying Algorithm

For the simulation, let there be three positions (A , B , C), and for those positions, three inter-distance measurements (L_{AB} , L_{BC} , L_{CA}) respectively as shown in Fig. 3. As the GNSS errors are more significant than the inter-distance measurement errors, if the distances between the GNSS positions do not match the inter-distance measurements, we should find a new set of coordinates (A' , B' , C') that does so.

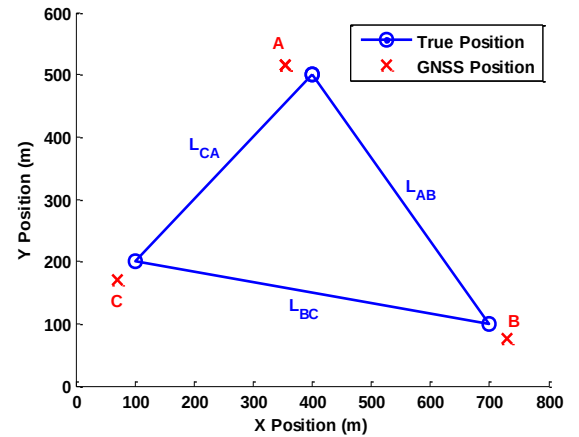


Fig. 3 Input Positions for the Simulation

First of all, if the distance between the points A and B (D_{AB}) is not equal to L_{AB} , we should find a new point A' such that the inter-distance error (E) between the

coordinate set (A', B, C) is minimized.

$$E = (D_{A'B} - L_{AB})^2 + (D_{BC} - L_{BC})^2 + (D_{CA'} - L_{CA})^2 \quad (14)$$

Afterwards, we update the GNSS position A with A' . Next, the GNSS position B' is determined in the same way.

$$E = (D_{A'B'} - L_{AB})^2 + (D_{B'C} - L_{BC})^2 + (D_{CA'} - L_{CA})^2 \quad (15)$$

After updating B with B' , we finally find C' also in the same light.

$$E = (D_{A'B'} - L_{AB})^2 + (D_{B'C'} - L_{BC})^2 + (D_{C'A'} - L_{CA})^2 \quad (16)$$

As a result, we are left with the coordinates (A', B', C') that satisfy the inter-distance measurements. From this set, we should determine a set of coordinates $(A^{(2)}, B^{(2)}, C^{(2)})$ that gives the minimum error from the true positions, which are defined as the positions with higher accuracy.

Therefore, we minimize the sum (S) of the distances between the set of coordinates and the initial GNSS positions. It is important to note that, in order to sustain the inter-distance measurements correspondence, the triangular shape created by the set (A', B', C') should be assumed to be an isometry. Therefore the only transformations allowed on the set of coordinates (A', B', C') are translation and rigid body rotation.

$$S = D_{A^{(2)}A} + D_{B^{(2)}B} + D_{C^{(2)}C} \quad (17)$$

Ultimately, the set of coordinates $(A^{(2)}, B^{(2)}, C^{(2)})$ gives more accurate positions than the initial set of coordinates (A, B, C) with respect to the true positions.

4. RESULTS

4.1 Error model validation

To validate the error model proposed in Sec. 2, and to test the rate of surveyed position convergence to the true position, a test was conducted using the data obtained from the DGNSS test-bed. Fig. 4 shows the locations of the DGNSS test-bed reference receivers within the KAIST campus. These three reference receiver antennas (shown in red) make up a configuration that is expected of a mobile DGNSS. The antennas are connected to three NovAtel OEM-4 reference receivers, which are in turn connected to a computer by a multiport serial board.

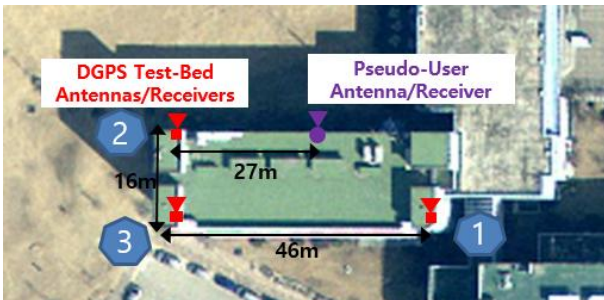


Fig. 4 Configuration of Mobile DGNSS Test-Bed at KAIST

Fig. 5 shows the surveyed position error of DGNSS test-bed receivers observed over eight consecutive days. In order to visualize when the surveyed position reaches a specified accuracy, the errors were calculated by finding the difference between the true position and the surveyed position which is the average of each of antenna positions (i.e., stand-alone GNSS solutions) collected up to that time. The positions were obtained from three stand-alone GNSS receivers. As the antennas are relatively close to one another, the antennas are affected by similar environmental errors such as multipath. Thus, the surveyed position errors show a similar trend over the given period. As shown in Fig. 5, at least one day worth of location measurements should be collected to achieve the position accuracy of less than 0.5 m error. The errors outlined in the figure are responsible for incurring erroneous differential corrections, which in turn increase the position errors experienced by the DGNSS user. The errors which the user experiences have been defined as the surveyed location error model in Sec. 2.

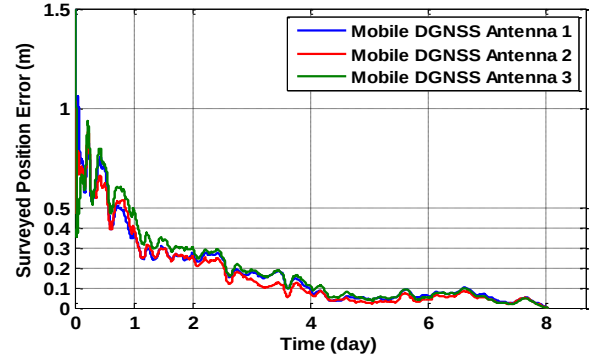


Fig. 5 Surveyed Position Error of DGNSS Test-bed Reference Receivers

To test the surveyed location error model of DGNSS, simulated errors were injected into the DGNSS test-bed. In this case, vertical error of three meters was injected into one of the three reference receivers. The Fig. 6 shows the vertical position solution of the DGNSS pseudo-user. The blue line represents the DGNSS vertical position error and the red line represents DGNSS vertical position error after three meters of surveyed location error injection. As shown in Fig. 6, after the surveyed position error injection, vertical error (the red line) increases by one meter. In Sec. 2, the DGNSS error induced by surveyed location error is defined as the sum of the surveyed location errors divided by the number of reference receivers. Therefore the result corresponds with the theoretical model outlined in Sec. 2.

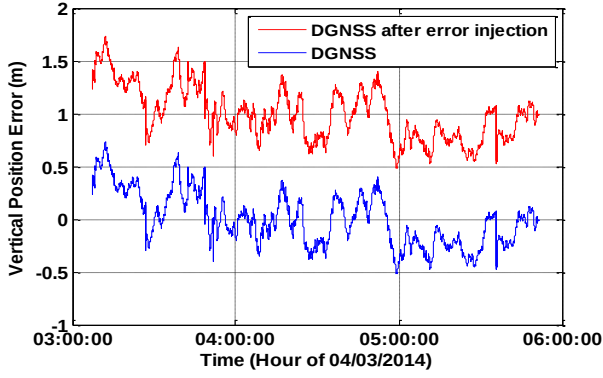


Fig. 6 DGNSS Vertical Position Errors before and after Surveyed Position Error Injection

4.2 Accuracy of relative distances

Using the data obtained from the DGNSS test-bed receivers, the relative distance estimation algorithm was validated. While the antennas of the two base stations were set to be stationary, L1/L2 code and carrier phase range data were collected from the DGNSS test-bed receivers.

Fig. 7 represents the accuracy of the relative distances between the two reference stations. The distance error was found to be completely bounded in the ± 1 cm level for the entire period. This confirms that the carrier phase double difference method can be used to provide relative distances of centimeter-level accuracy. Also, this confirms that the relative distance errors are not very significant compared to the GNSS surveyed position errors, which is the condition required for the fast location surveying algorithm.

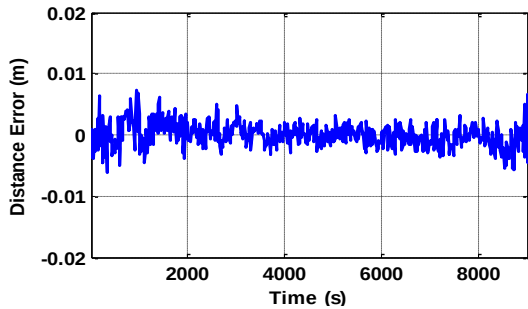


Fig. 7 Relative Distance Error between two stations

4.3 Fast location survey algorithm

To validate that the fast location surveying algorithm is effective in lowering the overall position error, a simulation was carried out. The simulation was carried out with low error correlation. To create the simulation data, for a given set of true positions, random errors of zero mean Gaussian distribution were injected into each of the coordinates. The positions with errors were set as the initial positions, and the positions that had gone through the algorithm were considered to be the processed positions.

For the simulation, 100 sets of initial positions with low error correlation were created from a true position and processed. The threshold for the quality check, which was mentioned in Section 3.1, was set so that the confidence level of the accepted data was at least 75%. Fig. 8 shows the error between the surveyed positions and the true position for each station when the error correlation is low. From the figure, it can be seen that the blue line, which represents the surveyed positions determined by the processed positions, approaches the 0 error faster than the red line, which is the initial positions' counterpart. This suggests that the processed positions allow the user to obtain accurate surveyed positions in a shorter amount of time.

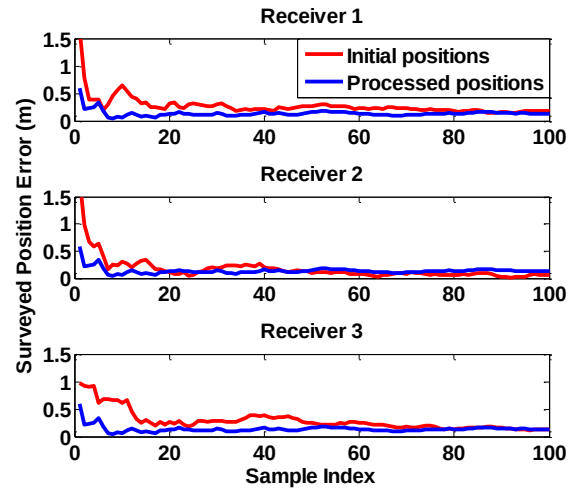


Fig. 8 Comparison of Position Errors for Initial Positions and Processed Positions (Low Correlation)

To check the effectiveness of the algorithm in realistic situations, we also tested the algorithm for the GNSS data collected from the DGNSS test-bed. The threshold was given as the confidence level of 75%. From Fig. 9 we can see that there has been some improvement, but the difference between positioning using initial positions and positioning using processed positions is not significant. The lack of improvement is due to the fact that the KAIST antennas are close in proximity and exposed to similar environments. From Fig. 4, we can see that the furthest distance between the receivers is approximately 48.7m and all the receivers are located in completely open space. Therefore, they are exposed to similar errors such as multipath as mentioned previously, which results in high correlation of GNSS errors. Consequently, as the errors are almost perfectly correlated, the performance of the algorithm is limited, as mentioned in Section 3.3.

It is important to note that in the case of actual battlefield environments where the mobile DGNSS will be operated at, there may be significant errors with low correlation. In those cases, the fast location surveying algorithm will most likely be highly effective.

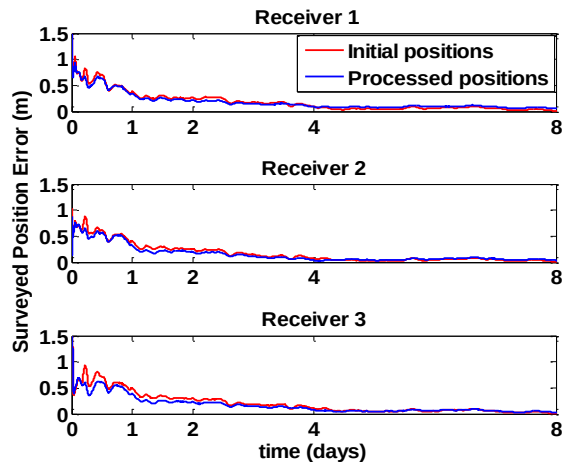


Fig. 9 Comparison of the Surveyed Position Errors for GNSS Initial Positions and Processed Positions

5. CONCLUSION

This paper proposes the theoretical model of DGNSS position errors induced by the inaccurate locations of the reference stations. By using the GNSS data collected from the DGNSS test-bed, the surveyed position's position accuracy of 0.5 m was found to be reached after at least one day. Also using these data, the proposed model was verified.

Furthermore in order to reduce the position errors in a short amount of time, a fast location surveying algorithm is suggested to decrease the time taken to determine surveyed positions of specific accuracy. The requirement of this algorithm is for the relative positions to be accurate. This paper presents how inter-distance measurements of such accuracy can be obtained from the carrier phase measurements taken from the GNSS receivers. This implies that no other instruments apart from GNSS sensors are required for the fast location surveying process.

Through the simulation tests, the fast location surveying algorithm is verified to be effective in obtaining more accurate positions than the raw GNSS positions. This will lead to a decrease in the time taken to achieve the desired accuracy level surveyed position. However, two things need to be considered. Firstly, the threshold for the algorithm influences the results significantly, as it determines which data should be discarded. Thus, appropriate threshold values must be assigned with respect to the accuracy of discrete GNSS measurements. Secondly, if the correlation of position errors of reference receivers is high, the algorithm's effectiveness is limited. Consequently, there needs to be further research so that significantly better performance can be attained despite the high correlation. Although this algorithm may have little influence on the GNSS position measurements made in conditions where the receivers experience errors of high correlation, in the unpredictable environments of the battlefield, the algorithm promises to be highly effective.

ACKNOWLEDGEMENTS

Dong-Kyeong Lee was supported by the Agency for Defense Development under the contract UE124026JD.

REFERENCES

- [1] J. Rife, S. Khanafseh, S. Pullen, D. Lorenzo, U. Kim, M. Koenig, T. Chiou, B. Kempny and B. Pervan, "Navigation, interface suppression, and fault monitoring in the sea-based joint precision approach and landing system," *Proceedings of the IEEE*, Vol. 96, No. 12, pp. 23–53, 2008.
- [2] M-B. Heo and B. Pervan, "Carrier phase navigation architecture for shipboard relative GPS," *IEEE Trans. Aero. and Elect. Sys.*, Vol. 42, No. 2, pp. 670–679, 2006.
- [3] P. J. G. Teunissen, "The least-squares ambiguity decorrelation adjustment: a method for fast gps integer ambiguity estimation," *Journal of Geodesy*, Vol. 70, Issue 1-2, pp. 65-82, 1995.
- [4] D. J. Paul, and T. Christian, "The LAMBDA method for integer ambiguity estimation: implementation aspects," *Technical Representative, LGR Series*, No. 12, Delft Geodetic Computing Centre, Delft University of Technology, The Netherlands, 1996.